

Numerical Derivatives — One-Page Summary

Finite Differences (first derivative)

Stencil	Formula	Trunc. err.	$h_\star$	$E_\star$
Forward	$\frac{f(x_0 + h) - f(x_0)}{h}$	$\mathcal{O}(h)$	$\sim \varepsilon^{1/2}$	$\sim \varepsilon^{1/2}$
Backward	$\frac{f(x_0) - f(x_0 - h)}{h}$	$\mathcal{O}(h)$	$\sim \varepsilon^{1/2}$	$\sim \varepsilon^{1/2}$
Central	$\frac{f(x_0 + h) - f(x_0 - h)}{2h}$	$\mathcal{O}(h^2)$	$\sim \varepsilon^{1/3}$	$\sim \varepsilon^{2/3}$

($\varepsilon \equiv \varepsilon_{\text{mach}}$; scalings assume $|f|, |f'|, \dots = \mathcal{O}(1)$.)

Why an optimal h exists. Total error = truncation + roundoff: $E(h) \approx C_{\text{trunc}} h^p + C_{\text{round}} \varepsilon/h$. The ε/h term comes from subtracting nearly equal f -values and dividing by h . Too large h : truncation dominates. Too small h : cancellation dominates. Minimising gives $h_\star \propto \varepsilon^{1/(p+1)}$.

Higher Derivatives and Non-Uniform Grids

Second derivative (uniform grid): $f''(x) \approx [f(x+h) - 2f(x) + f(x-h)]/h^2 + \mathcal{O}(h^2)$. Optimal $h_\star \sim \varepsilon^{1/4}$.

Non-uniform first derivative ($h_1 = x_i - x_{i-1}$, $h_2 = x_{i+1} - x_i$): quadratic fit through three points $\Rightarrow \mathcal{O}(h^2)$; reduces to central difference when $h_1 = h_2$.

Partial / mixed derivatives: apply 1-D stencils along each axis. E.g. $\partial_{xy} f \approx [f(x+h_x, y+h_y) - f(x-h_x, y+h_y) - f(x+h_x, y-h_y) + f(x-h_x, y-h_y)]/(4h_x h_y)$.

Complex-Step Derivative

For analytic f , evaluate at $x + ih$:

$$f'(x) = \frac{\text{Im}[f(x + ih)]}{h} + \mathcal{O}(h^2).$$

No subtraction of close numbers $\Rightarrow$ no cancellation $\Rightarrow$ can take $h \sim 10^{-20}$ or smaller safely. Gives nearly machine-precision derivatives with $\mathcal{O}(h^2)$ truncation error. Requires f to be implementable for complex arguments.

Automatic Differentiation (AD)

AD differentiates the *program itself* by attaching derivative info to every elementary operation via the chain rule.

	Forward mode	Reverse mode
One sweep gives	1 directional derivative	full gradient ∇f
Cost / sweep	$\sim 1 \times$ function eval	$\sim 2\text{--}5 \times$ function eval
Memory	low	stores intermediate tape
Best when	few inputs	many inputs, scalar output

Forward mode: attach dual numbers $(v, \dot{v})$ to each variable; seed $\dot{x} = 1$ for the chosen input; propagate with local rules (e.g. $(a, \dot{a})(b, \dot{b}) = (ab, \dot{a}b + a\dot{b})$).

Reverse mode: run forward (record tape), then propagate adjoints $\bar{u}_j = \sum_i \bar{u}_i \partial u_i / \partial u_j$ backward from $\bar{f} = 1$ to the inputs.

Precision: same order as evaluating f itself (no differencing artifacts). Cannot fix ill-conditioned computations.

Convergence and Self-Convergence

Known solution: for a method of order n , $Q = |y_{h_2} - y| / |y_{h_1} - y| \approx (h_2/h_1)^n$.

Unknown solution (self-convergence, three resolutions $h_1 > h_2 > h_3$):

$$Q_S = \frac{\|y_{h_3} - y_{h_2}\|}{\|y_{h_2} - y_{h_1}\|} \approx \frac{h_3^n - h_2^n}{h_2^n - h_1^n}.$$

Which Method When?

Method	Use case
Finite differences	Quick estimates; f is a black box; few derivatives needed
Complex step	Near-exact first derivatives; f extends to $\mathbb{C}$; no cancellation worries
AD (forward)	Few inputs; exact derivatives of coded functions; Jacobian–vector products
AD (reverse)	Scalar loss with many parameters (optimisation, HMC); full gradient in one pass

Bottom line: finite differences are simple but limited by accuracy floors (scaling $\varepsilon^{1/2}$ or $\varepsilon^{2/3}$ for first derivative) and scale poorly with dimension. Complex step removes the cancellation barrier. AD removes both the cancellation barrier *and* the dimensional cost, at the price of tooling.